Module 4: Evaluation

Evaluator Introduction

Evaluator? I hardly know her!

What are we focused on today?

1. How does the Visitor pattern works for evaluation?
2. How are language values represented at runtime?
3. How does our implementation affect evaluation order?

1.

Tree-Walk Interpreter

“The Ents are marching on Isengard!”

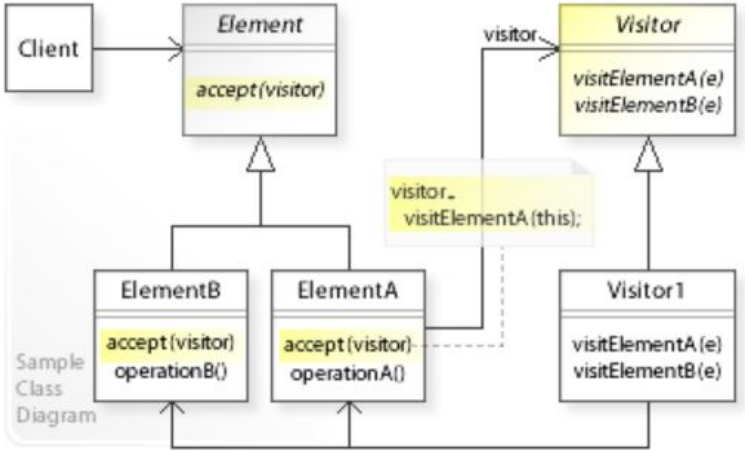
How do we evaluate AST classes?

We need some kind of “.evaluate()” method on AST…
…but it would be best not to modify the AST classes.

Visitor Pattern

The visitor pattern allows defining new operations on an existing class hierarchy.

- The Visitor class defines visitX methods, implementing the operation for class X.
  - The accept method on class X calls the corresponding visitX method.
- Effectively, this adds polymorphic behavior to existing classes!
- Another variant is to use instanceof checks instead of an accept methods.
  - Sealed classes help type safety here!



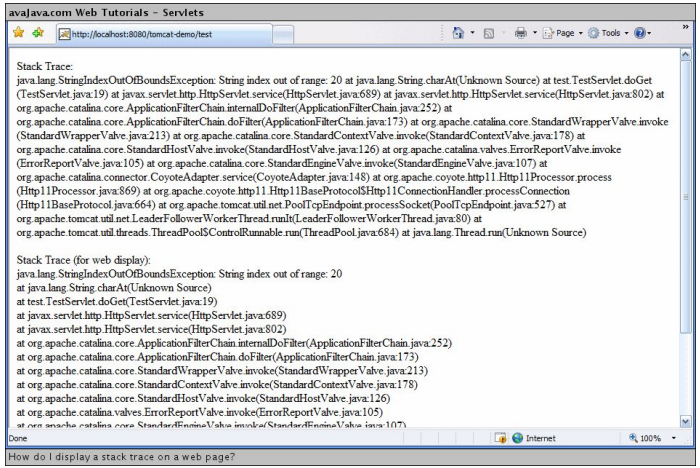
```java
sealed interface Ast {
  record Number(...) implements Ast {}
  record Variable(...) implements Ast {}
  record Function(...) implements Ast {}
}

public BigDecimal visit(Ast ast) throws CalculateException {
  return switch (ast) {
    case Ast.Number number -> visit(number);
    case Ast.Variable variable -> visit(variable);
    case Ast.Function function -> visit(function);
  };
}
public BigDecimal visit(Ast.Number ast) { … }
public BigDecimal visit(Ast.Variable ast) { … }
public BigDecimal visit(Ast.Function ast) { … }
```

Visitor Examples

Runtime Errors

Errors during execution are **runtime** errors, which will typically cause abandonment.

- Our language is dynamic and interpreted, meaning most errors are at *runtime*.
  - Incorrect types, e.g. `1 + true`.
  - Undefined variables/functions.
  - “Real” runtime errors, e.g. `1/0`.
- Statically typed, compiled languages often have runtime checks as well.
- Errors in userland are *not* runtime errors!
  - These are part of the user’s program!



2.

Evaluation Time

Not exams of course - I'm not that cruel.

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